

Polysemy—where single word forms possess multiple related meanings (e.g., "paper" as material, document, or newspaper)—presents a fundamental challenge for theories of semantic representation and processing. Unlike *homonymy* (with unrelated meanings, e.g., "bat" as an animal vs. as sports equipment), polysemous senses exhibit semantic relationships that computational approaches can now quantify. For example, BERT, a state-of-the-art neural language model, produces a contextualized vector representation of a word's sense in continuous semantic space based on the surrounding words in a sentence. *BERT cosine distance* is a metric that indicates the semantic dissimilarity between contextualized meanings of a polysemous word (e.g., for "marinated lamb" and "friendly lamb"). Theoretical accounts of polysemy, however, diverge sharply: *does polysemy resolution involve selection between senses with discrete categorical boundaries, continuous modulation along semantic gradients, or a hybrid architecture?* Behavioral studies using primed sensibility judgments show that both BERT distance and dictionary-defined sense boundaries independently predict processing difficulty (Trott & Bergen, 2023). Yet, these factors naturally covary in language, making definitive adjudication of theories difficult. Neural validation is essential but lacking (Pylkkänen et al., 2006): No study has orthogonally manipulated categorical boundaries and continuous distance while measuring *lexical access* (M350, 250-450ms), *semantic integration* (N400, 300-500ms), and *cognitive control* (theta oscillations, 400-800ms). This multi-stage, multi-measure approach is necessary because competing accounts of polysemy make distinct predictions across processing stages—predictions that remain untested.

The objective of this study is to determine which theoretical framework best accounts for the neural dynamics of polysemy processing. **Our central hypothesis** is that categorical sense boundaries and continuous semantic distance operate through stage-specific neural mechanisms: sense boundaries modulate early lexical access (M350 latency), continuous distance governs semantic integration difficulty (N400 amplitude), and boundary crossing triggers theta-band cognitive control and amplifies the effect of semantic distance.

Specific Aim 1: Adjudicate among competing models through multi-stage neural measures. Using magnetoencephalography (MEG) during primed sensibility judgments, we will orthogonally manipulate continuous semantic distance and categorical boundaries ("same-sense" vs. "different-sense" pairs), measuring: (1) M350 peak latency in the left superior temporal cortex, (2) N400 mean amplitude in centro-parietal sensors, and (3) theta power (4-8Hz) in left frontal-temporal regions. Formal model comparison (AIC/BIC) will evaluate Pure Categorical, Pure Continuous, Additive Hybrid, and Interactive Hybrid models. *Hypothesis 1a:* M350 latency will show primarily categorical effects, with facilitated latencies for same-sense versus different-sense pairs, reflecting shared lexical access for related senses (Pylkkänen et al., 2006).

Hypothesis 1b: N400 amplitude will show primarily continuous distance effects, with more negative amplitudes for larger semantic distances, regardless of sense boundaries (Klepousniotou et al., 2008). *Hypothesis 1c:* Theta power will show both categorical and distance effects, with increased control for different-sense pairs and larger distances (Bastiaansen et al., 2008).

Specific Aim 2: Determine whether frontal-temporal coordination differs for continuous versus categorical mechanisms. We will measure phase-amplitude coupling (PAC) between frontal theta phase and temporal alpha amplitude (8-12Hz). *Hypothesis 2:* Theta-alpha PAC will show stronger coupling for different-sense versus same-sense pairs, indicating enhanced frontal-temporal coordination when crossing boundaries. This study will provide the first millisecond-resolution neural evidence adjudicating among competing theories of polysemy processing. If the Interactive Hybrid model best fits data across measures, this would reconcile decades of theoretical debate by demonstrating that categorical and continuous mechanisms operate through distinct but interacting neural substrates. The findings will advance our fundamental understanding of how the brain constructs meaning from ambiguous linguistic input, with broader applications to

¹ Note: In this proposal, we will be using the terms "meaning" and "sense" interchangeably. Senses are the distinct interpretations of a word that emerge in different contexts. Practically speaking, a word sense is a particular meaning that can be listed in a dictionary as a separate entry or sub-entry.

computational models of language, information retrieval systems, and clinical interventions for semantic processing deficits.

Background and Significance

The Polysemy Processing Debate

Lexical ambiguity pervades natural language: 84% of dictionary entries contain multiple senses, with 37% exhibiting five or more distinct meanings (Rodd et al., 2004). Polysemy occurs when these meanings are semantically related—"paper" can refer to a material, an academic article, a newspaper, or an assignment. Understanding how the brain resolves such ambiguity clarifies semantic representation and processing, yet the cognitive architecture underlying this process remains contested.

Three theoretical frameworks make distinct predictions. The **Mental Dictionary Framework** posits that each polysemous word stores multiple discrete sense definitions, with context triggering categorical selection (Elman, 2009). This predicts facilitation when accessing the same sense consecutively and inhibition when switching senses, with minimal graded variation within boundaries—supported by priming studies showing sense-specific effects (Klein & Murphy, 2001). Our first model, **Pure Categorical**, tests this account.

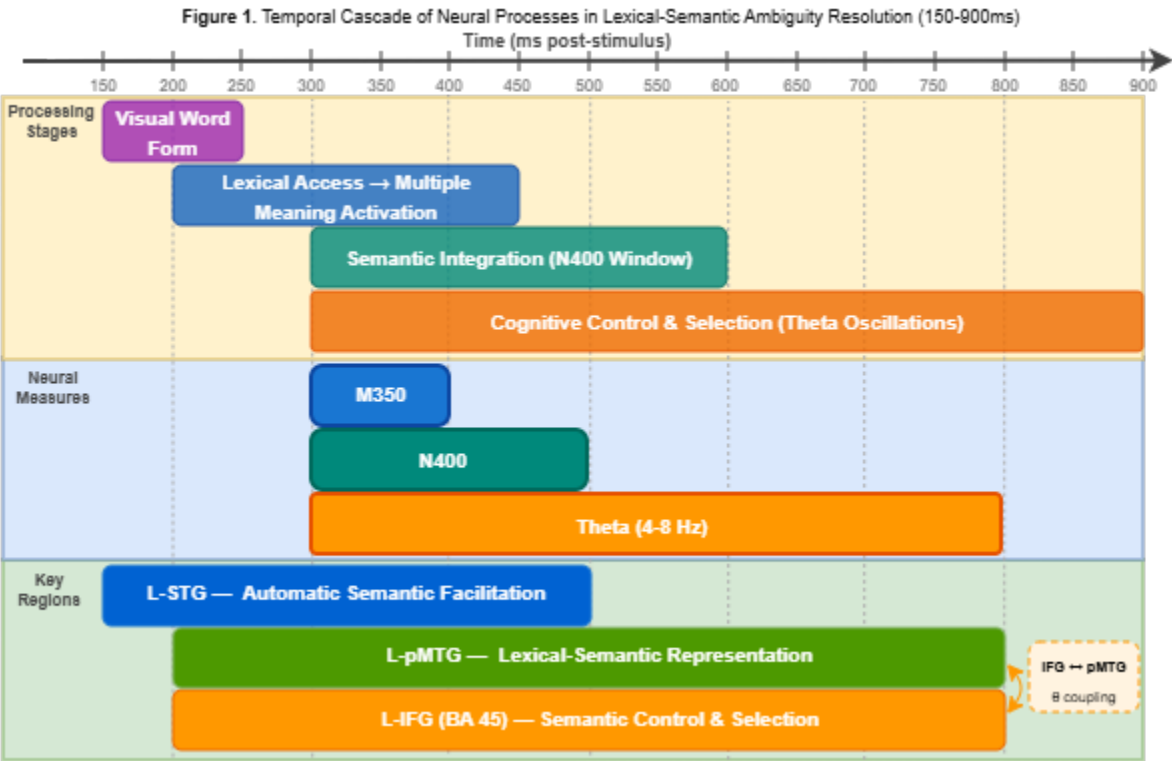
The **Continuity of Meaning Framework** rejects discrete senses entirely. Li & Joanisse's (2021) computational model treats senses as clusters emerging from contextual co-occurrence, without distinct boundaries.

Processing difficulty scales continuously with semantic distance regardless of dictionary classifications. Our second model, **Pure Continuous**, predicts gradient effects without discontinuous jumps at sense boundaries.

The **Hybrid Meaning Framework** integrates both perspectives. Trott & Bergen (2023) demonstrated that continuous semantic distance and categorical sense boundaries independently predict behavioral processing difficulty. Our third model (**Additive Hybrid**) tests whether distance and boundaries contribute independently, while our fourth (**Interactive Hybrid**) tests whether crossing boundaries amplifies distance effects.

Neural Evidence and Critical Gaps

Behavioral studies cannot reveal how or when these mechanisms operate. Neural evidence reveals distinct processing stages (Figure 1), but it has not tested orthogonally manipulated predictors across those stages.



M350 (300-400ms) indexes lexical access in the left superior temporal cortex (Pylkkänen & Marantz, 2003). M350 latency shows categorical sensitivity: polysemous senses sharing a lexical entry produce facilitation (earlier peaks ~337ms), while homonymous meanings in separate entries produce inhibition (delayed peaks ~361ms), even when semantic distances are matched (Beretta et al., 2005; Pylkkänen et al., 2006). However, existing studies confound categorical boundaries with semantic distance, leaving open the question of whether M350 also shows graded effects within categories. If a hybrid architecture operates during lexical access, M350 should show both categorical and continuous effects.

N400 (300-500ms) reflects semantic integration difficulty (Kutas & Hillyard, 1980). Klepousniotou et al. (2008) demonstrated that processing difficulty scales continuously from metonymic polysemy (high overlap, minimal cost) through metaphorical polysemy (moderate overlap) to homonymy (low overlap, maximal cost). N400 amplitude tracks this gradient while showing qualitatively distinct dynamics—metonymy shows sustained priming, homonymy shows rapid decay (MacGregor et al., 2015). This dual sensitivity indicates a hybrid architecture where categorical organization and continuous similarity coexist (Trott & Bergen, 2023), yet no studies have orthogonally manipulated these factors.

Theta oscillations (4-8Hz) in left frontal regions index cognitive control during ambiguity resolution, with increased power reflecting selection demands when meanings compete (Bastiaansen et al., 2008). **Theta-alpha phase-amplitude coupling (PAC)** between left inferior frontal gyrus and posterior middle temporal gyrus reveals how frontal control coordinates semantic retrieval—theta phase modulates alpha amplitude, quantified via the Modulation Index (Tort et al., 2010). If categorical boundaries trigger distinct control processes, PAC should increase for different-sense pairs; if control scales continuously, PAC should correlate with semantic distance regardless of boundaries. No studies have tested these alternatives.

Critical Gap: Trott & Bergen (2023) showed both distance and boundaries independently predict behavior, but neural studies confound these factors. The present study orthogonally manipulates both while measuring M350 (lexical access), N400 (semantic integration), and theta-alpha coupling (cognitive control) to test whether the hybrid architecture reflects stage-specific processing or parallel operation across all stages.

Innovation. Our multi-stage measurement captures the temporal cascade of meaning selection across lexical access, semantic integration, and cognitive control that single-measure studies cannot dissociate. A fundamental limitation of prior polysemy research is the natural collinearity between semantic distance and sense boundaries: same-sense word uses occur in more similar contexts than different-sense uses. Our orthogonal design eliminates this shared variance, enabling a definitive causal test of whether continuous distance and categorical boundaries independently affect neural processing. Formal model comparison determines which theoretical framework best explains neural dynamics.

Clinical Significance. Semantic aphasia patients struggle with polysemous words. Establishing whether deficits reflect categorical selection failure, continuous distance insensitivity, or impaired control coordination would inform targeted interventions.

Broader Impact. Polysemy processing exemplifies a fundamental cognitive tension: balancing flexibility with structure. Like color perception (continuous wavelengths perceived as discrete categories) and speech perception (continuous acoustic phenomena perceived as phonemes), semantic processing may impose categorical boundaries that accentuate between-category differences while minimizing within-category variation (Goldstone & Hendrickson, 2010). Validating the Hybrid Meaning Framework would illuminate how discrete structure emerges from continuous representations across cognitive domains.

Research Design and Methods

Overview and Experimental Design

This study employs a **within-subjects design using MEG** to test whether neural dynamics during polysemous word comprehension reflect the Continuity of Meaning Framework (Li & Joanisse, 2021), the Mental Dictionary Framework (Elman, 2009), or the Hybrid Meaning Framework (Trott & Bergen, 2023). Following Trott & Bergen (2023), we will vary two predictors of processing difficulty. **Continuous semantic distance** will be measured as the BERT cosine distance (BERT_distance) between vector representations in the continuous semantic space of a word's sense in the prime and target sentences. BERT distances inversely capture distributional semantic similarity. Smaller distances indicate more similar meanings (e.g., "chicken" as bird vs. "chicken" as coward). Larger distances indicate less similar meanings (e.g., "book" as physical object vs. its informational content). **Categorical sense boundary** is a binary variable indicating whether prime and target contexts activate the same dictionary-defined sense (sense_boundary = 0) or different senses (sense_boundary = 1) of the polysemous word. In other words, sense_boundary = 1 indicates that a categorical boundary exists between vector representations of a word's sense in the prime and target sentences. **Orthogonality.** A critical methodological limitation in polysemy research is the natural collinearity between semantic distance and sense boundaries: same-sense pairs cluster at high relatedness (M = 3.53) while different-sense pairs cluster at low relatedness (M = 1.38; Trott & Bergen, 2023). Although Trott & Bergen demonstrated that both predictors explain independent variance, the inherent correlation prevents definitive causal conclusions about whether each factor independently affects processing. Our design (Table 1) eliminates this confound by oversampling naturally rare combinations (e.g., same-sense pairs at high BERT distances and different-sense pairs at low distances), achieving statistical independence (target $r < 0.15$). By addressing the primary limitation of prior behavioral research, our design provides the definitive neural validation needed to adjudicate between competing theoretical frameworks.

Table 1

Conditions resulting from orthogonally-varied categorical boundaries and continuous distance

	Low distance	High distance
Same-sense	Condition: Close same-sense (Baseline) Prediction: minimal processing load Ex: "excellent book" vs. "outstanding book"	Condition: Distant same-sense Validates Pure Continuous model if: continuous distance effect without boundary crossing effect Ex: "excellent book" vs. "mediocre book"
Different-sense	Condition: Close different-sense Validates Pure Categorical model if: categorical boundary effects with minimal semantic distance Ex: "excellent book" vs. "pristine book"	Condition: Distant different-sense Validates Hybrid model if: combined effects of both distance and boundaries Ex: "excellent book" vs. "tattered book"

Participants

[Sample size] right-handed native English speakers [age range] will be recruited from the university and local community. Sample size provides [power analysis here]. Exclusion criteria include neurological disorders, psychiatric medications, metal implants, and left-handedness (to ensure left-hemisphere language dominance). All participants will provide informed consent in accordance with IRB-approved protocols.

Stimuli and Procedure

Stimulus Development. Following Trott & Bergen (2023), we select 120 polysemous words with clear dictionary-defined sense distinctions (using the Merriam-Webster and Oxford English dictionaries). For each word, we create four prime-target sentence pairs: close same-sense, distant same-sense, close different-sense, and distant different-sense. Each sentence pair contains an overlapping word (e.g., "lamb") and same sentence frame (e.g., "They liked the ___ lamb"), but differ in the disambiguating word (e.g., "marinated" vs.

“friendly”). Similarly, we select 120 “filler” (non-ambiguous) words. Each filler word is matched for the concreteness, frequency, part of speech, and number of syllables of one of the polysemous words. Two sentences are constructed for each filler word. For 40 of these filler words, both sentences are nonsensical; for the remaining 80, only one of the two sentences is nonsensical (counterbalanced for whether the first or second is nonsensical). This is to prevent participants from learning any contingencies between a word in the priming and target sentences. Contextualized vector representations of the critical word in each sentence pair are extracted from BERT, and cosine distances are computed between them.

Experimental Task. Participants will perform a primed sensibility judgment task. Participants are told they will read a series of sentences, and that some will make sense while others will not. Their task is to assess whether each is sensical; they are told to indicate this by pressing a button with their left or right hand. Participants are encouraged to complete each trial as quickly and accurately as possible. Before beginning the primary experiment, participants complete ten practice trials (five sentence pairs). Sixty words are randomly sampled for the primary experiment from the set of 240 (polysemous and non-ambiguous) words; then, for each polysemous word sampled, one of the four corresponding sentence pairs is sampled. On each trial, a participant sees a prime-target sentence pair appear consecutively in the center of the monitor (e.g., “They liked the marinated lamb” followed by “They liked the friendly lamb”). Prime and target sentences appear on different pages, with an intertrial interval of 0ms. Trial order is pseudorandomized with constraints: (1) balanced condition distribution (Table 1), (2) < 25% of trials with nonsensical sentences. Total session time will be [X.X] hours, including setup and calibration.

MEG Data Acquisition and Preprocessing

Data Acquisition. [technical details]. **Preprocessing Pipeline.** [technical details]

Neural Measures and Statistical Analyses

Step 1: M350 Latency Analysis (Lexical Access). M350 peak latency will be extracted from left superior temporal sensors within a 300-400ms post-target-onset window. Peak latency is defined as the time point of maximum magnetic field strength within this window. This measure indexes the speed of lexical entry access. We will fit four competing mixed-effects models (using `lme4` in R) to M350 latencies (following Trott & Bergen, prior_latency is a control variable, which accounts for trial-order effects):

Model 1 (Pure Categorical): $M350 \sim \beta_0 + \beta_1(\text{sense_boundary}) + \beta_2(\text{prior_latency}) + (1 | \text{participant}) + (1 | \text{word})$

Model 2 (Pure Continuous): $M350 \sim \beta_0 + \beta_1(\text{BERT_distance}) + \beta_2(\text{prior_latency}) + (1 | \text{participant}) + (1 | \text{word})$

Model 3 (Additive Hybrid): $M350 \sim \beta_0 + \beta_1(\text{BERT_distance}) + \beta_2(\text{sense_boundary}) + \beta_3(\text{prior_latency}) + (1 | \text{participant}) + (1 | \text{word})$

Model 4 (Interactive Hybrid): $M350 \sim \beta_0 + \beta_1(\text{BERT_distance}) + \beta_2(\text{sense_boundary}) + \beta_3(\text{BERT_distance} \times \text{sense_boundary}) + \beta_4(\text{prior_latency}) + (1 | \text{participant}) + (1 | \text{word})$

Model comparison will use Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), with $\Delta\text{AIC} > 10$ indicating strong evidence for model superiority (Anderson & Burnham, 2002).

Step 2: N400 Amplitude Analysis (Semantic Integration). N400 mean amplitude will be extracted from centro-parietal sensors within the 350- 500ms post-target window. We will fit the same four competing models above to N400 amplitudes.

Step 3: Theta Power Analysis (Cognitive Control). Theta power change (relative to baseline) will be computed for the 300-800ms post-target window. The same four models will be fit to theta power.

Step 4: Phase-Amplitude Coupling (Network Coordination). For Aim 2, we will compute theta-alpha phase-amplitude coupling (PAC) between left frontal theta phase (4-8Hz) and left temporal alpha amplitude (8-12Hz) using the Modulation Index (Tort et al., 2010). PAC will be measured in sliding windows from 200 to 600ms post-target. We will test whether PAC: (1) differs between same-sense and different-sense conditions (categorical effect), and (2) correlates with BERT_distance within same-sense trials (continuous effect).

Predictions and Interpretation of Results

If a model is the best fit for all three stages, we can definitively say that the corresponding framework is the mechanism of lexical-semantic ambiguity resolution. For example, if model 2 is the best fit in all three stages,

we validate Li & Joanisse’s (2021) continuous modulation theory. Heterogeneous results (i.e., different models are the best fit for the three stages) require unique interpretations. Our central hypothesis is that sense boundaries modulate early lexical access (Model 1 is the best fit for M350 latency), continuous distance governs semantic integration difficulty (Model 2 is the best fit for N400 amplitude), and boundary crossing triggers theta-band cognitive control and amplifies the effect of semantic distance (Model 4 is the best fit for theta power change). Table 2 provides information on our central hypothesis and the related scenario in which Model 3 offers the best fit for theta power change.

Table 2
Scenarios with Corresponding Theoretical Conclusion and Interpretation

Stage 1 (M350)	Stage 2 (N400)	Stage 3 (Theta)	Theoretical Conclusion	Mechanistic Interpretation
1	2	3	Categorical Access, Continuous Integration, Additive Control	Lexical access relies on discrete sense competition; integration operates on continuous semantic similarity; cognitive control tracks both independently (boundary crossing triggers selection; distance modulates retrieval effort). Representational format transitions from symbolic/discrete to distributed/continuous across stages.
1	2	4	Categorical Access, Continuous Integration, Interactive Control	Cognitive control exhibits compound architecture; boundaries amplify distance-driven control demands, suggesting executive monitoring integrates both discrete structural violations and continuous retrieval difficulty through an interactive mechanism. Control systems detect when boundary crossing co-occurs with large semantic distance, recruiting maximal executive resources.

Potential Problems and Alternative Strategies

- Problem 1: Insufficient orthogonality between distance and boundaries.** If natural language constraints prevent entirely orthogonal stimulus design (e.g., correlation $r > 0.3$), we will include the correlation as a covariate in our models and interpret results cautiously regarding independence.
- Problem 2: No model clearly wins in a given stage.** Interpretation: Multiple theoretical accounts remain plausible; data insufficient to adjudicate. Response: Report Akaike weights quantifying relative model probabilities; employ multimodel inference (Burnham et al., 2011), averaging parameter estimates across competing models to acknowledge model-selection uncertainty.
- Problem 3: Stage-specific effects only** (e.g., M350 shows effects but N400 and theta do not). Interpretation: Critical processing occurs within a narrow time window; effects are resolved before later stages. Mechanistic insight: Identifies specific computational bottleneck. For example, if only M350 is affected, polysemy resolution is fundamentally a lexical-access problem rather than an integration or control challenge.
- Problem 4: Weak or absent theta effects.** If theta power does not differentiate between conditions, we will examine alternative control measures, such as frontal-temporal connectivity, using coherence measures. Our rationale is that multiple neural signatures of cognitive control exist; theta is predicted, but not the only possible index (Lewis & Bastiaansen, 2015).

Timeline

Months 1-6: Preparation Phase. [details] **Months 7-12:** Data Collection Phase. [details] **Months 13-16:** Analysis and Dissemination Phase. [details]

Resources and Institutional Support. MEG facility access is secured through the Department of Neurology at the University of Colorado Anschutz Medical Campus.

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